Meaning Before Naming

A Readout-Retention Architecture of Affective-Semantic Reorganization

Programme extension from Human Learning, Human LoRA, Readout Genesis, and Epistemic Fusion/Tunnel

Yaoharee Lahtee | Bangkok, Thailand | Standalone Concept Note v1.0 FULL

Status. K0 conceptual architecture and programme-extension paper. The paper does not propose a new theory of music, a universal code of emotion, or a literal physical theory of resonance. Music appears only as a phenomenological probe that makes a broader claim visible: meaningful human reorganization can occur before explicit naming, while the transmitted signal remains non-identical to the meaning that emerges in a particular reader. The architecture inherits its level distinctions from Readout Genesis, its embodied and affectively weighted account of experience from *Experience Is the Human LoRA*, its lived-word and semantic-network commitments from *Human Learning as Epistemic Architecture*, and its reader-, context-, and history-sensitive transport model from *Epistemic Fusion or Tunnel*. No dedicated review of music cognition is claimed here; any feature-specific account of melody, rhythm, timbre, harmony, or culture would require separate empirical review.

Central claim. Meaning need not begin with a name. A finite human can undergo an affectively weighted, context- and history-dependent reorganization in response to an external pattern before that change is propositionally articulated. The pattern does not carry a finished meaning from one interior to another. Rather, it supplies retained differences that a situated reader makes accessible through an existing and revisable organization of memory, embodiment, expectation, language, value, and context. Some such reorganizations vanish; others are retained and alter how later situations can be read, named, and acted upon.

1. Abstract

The Human-AI Readout Programme has treated language as epistemic mediation: articulation makes distinctions nameable, comparable, transportable, and revisable, while never becoming identical to the full human state from which it was produced [17]. Readout Genesis places retained difference before semantic naming and defines meaning as a context-, memory-, criterion-, and reader-dependent readout relation rather than a property stored in the root [16]. *Human Learning as Epistemic Architecture* begins from lived words whose meanings are entangled with experience, emotion, social origin, value, goal, evidence, and action [14]. *Experience Is the Human LoRA* further argues that lived experience is embodied, affectively weighted, context-mediated, and capable of leaving selectively retained residues that modify later interpretation and action [15]. These commitments together generate a residual question: must meaningful reorganization wait until experience has been explicitly named?

This paper answers no. It proposes a readout-retention architecture in which externally available differences can produce a reader-relative affective-semantic configuration before explicit lexical or propositional articulation. The central non-collapse is that *signal is not meaning, meaning is not naming, intensity is not warrant, and retention is not improvement*. Music is used only as a phenomenological probe: an acoustic pattern can be heard, felt, and experienced as uplifting, sorrowful, tense, peaceful, or energizing before the listener can state a proposition that exhausts what has changed. This does not imply that the sound contains a universal emotional code or that producer and listener share an identical inner state. It illustrates a more general structure: external difference → finite readout → affective-semantic reorganization → optional naming → selective retention → altered future readout.

The contribution is an extension of the programme below the level of explicit Human Return. It clarifies that the lived word is an important entry unit for reflective learning but not the ontological origin of all meaning; language is one powerful articulation and transport layer within a broader readout ecology. The paper specifies claim boundaries, failure modes, a minimal formal architecture, and an empirical programme for testing whether prenominative reorganization adds explanatory value beyond simpler accounts of arousal, verbal labeling, or memory.

2. The Residual Problem in the Existing Programme

The programme already contains most of the ingredients needed for this extension. The missing step is their placement.

Human Learning as Epistemic Architecture deliberately begins with the lived word rather than abstract information. A word is treated not as an inert dictionary unit but as a site where experience, memory, emotion, social pressure, value, goal, constraint, evidence, and action converge. Learning then moves from representation to relation, retrieval, evaluation, revision, problem-context, tool use, workflow, skill, and feedback [14]. This is a methodologically powerful starting point for education because words can be mapped, contested, sourced, and revised.

Yet Readout Genesis states something more basic: retained difference precedes the name of what differs, and meaning is attached only after a valid domain translation and readout. A reader makes a distinction accessible under a question; the reader does not retroactively place that meaning in the root [16]. If this architecture is taken seriously, the lived word cannot also be the universal first occurrence of meaning. It is an entry point for inquiry, not the metaphysical beginning of experience.

Experience Is the Human LoRA supplies the second pressure. Experience is described as a temporally thick, embodied, affectively weighted, and context-mediated readout through which a world encounter becomes present to a finite observer. The durable consequence of experience is not the full event but a selectively retained alteration in later interpretive and action dispositions [15]. A large portion of experience may remain transient; some residues persist without becoming detailed episodic memory; some may become habits, sensitivities, or skills. Thus explicit verbal recall cannot exhaust what counts as a meaningful retained change.

Finally, *Epistemic Fusion or Tunnel* treats articulation as a bounded export of a larger retained human state. Language makes distinctions available for transformation, but the utterance is not identical to the human state that generated it. The same programme also makes reader, context, history, provenance, and resistance part of the trajectory rather than treating meaning as a fixed property of a surface form [17, 18].

The residual problem is therefore narrow:

> *How can the architecture represent meaningful, affectively weighted human reorganization that occurs before explicit naming, without treating meaning as a substance encoded inside an external signal or collapsing felt change into truth?*

3. Internal Genealogy and What Must Not Be Rewritten

The present note is an extension, not a replacement. Table 1 fixes what is inherited and what remains new.

The key correction is a level distinction. **Words can be methodologically first without being phenomenologically first.** Human Learning can begin an educational protocol with a word because a word is inspectable and socially contestable. Readout Genesis can simultaneously maintain that the retained distinctions and readout relations from which meaning emerges are prior to semantic naming. There is no contradiction once method and ontology are kept separate.

4. Phenomenological Ground: Meaningful Change Can Precede Naming

The architecture begins from a small descriptive floor rather than from equations.

P1 - Finite access. A human never receives the totality of an event. What becomes present is selected through bodily capacities, attention, prior learning, memory, expectation, language, and context [4, 13, 11, 15].

P2 - Affective weighting. What becomes accessible is not affectively neutral. Experience can appear as inviting, threatening, consoling, beautiful, tense, familiar, alien, hopeful, or otherwise action-relevant before the person can provide a complete propositional account of why [15].

P3 - Relational meaning. The significance of a signal is not exhausted by its isolated surface form. It depends on active relations among prior experience, contextual frames, memories, values, expectations, and other accessible distinctions [5, 8, 14].

P4 - Naming can lag reorganization. A change in what feels salient, connected, possible, or action-relevant may occur before the person can name the distinction that changed. Explicit articulation can follow, stabilize, sharpen, distort, or fail to capture the prior reorganization.

Consider an ordinary but theoretically revealing case. A composer or performer may externalize part of an inner organization through patterned sound, yet what crosses the physical medium is not a finished sadness, hope, memory, or spiritual state. What is available to the listener is an acoustic pattern containing differences of timing, pitch, intensity, timbre, repetition, pause, and relation. A listener may then experience the pattern as deeply uplifting, sorrowful, peaceful, destabilizing, or energizing before being able to say what the experience "means." Another listener may hear the same performance and undergo a different change, or almost none. The point is not that music possesses a special metaphysical power. The point is that the example makes a general architecture visible: an external pattern can reorganize an internal field without transporting an identical inner state from one person into another. The felt "echo inside" is therefore better modeled as reader-relative reorganization than as semantic copying.

This example also exposes why the paper cannot be reduced to a theory of verbal semantics. If meaning were identical to explicit linguistic definition, then meaningful change prior to naming would be incoherent by definition. The programme's own prior commitments resist that collapse: Human LoRA already treats lived experience as embodied and affectively weighted, and Readout Genesis explicitly places meaning after readout rather than after lexical naming [15, 16].

5. The Core Readout-Retention Architecture

Let x_n denote a finite externally available pattern at step n. It can be linguistic, acoustic, visual, gestural, social, or multimodal. The present note does not

Table 1: Programme genealogy for the present extension. Internal work establishes continuity, not independent empirical validation.

Prior programme object	Retained result	Role here
Human Learning as Epistemic Architecture	Lived words are loaded by experience, memory, emotion, value, evidence, goals, constraints, and action; learning builds meaning neighborhoods, concept relations, problem-context, tools, workflow, skill, and feedback.	Supplies the reflective and educational level. The lived word remains an excellent unit for inquiry, but is retyped as a later articulation/access point rather than the origin of all meaning.
Experience Is the Human LoRA	Experience is embodied, affectively weighted, context-mediated, and not automatically learning; selected residues may persist and alter later interpretation/action.	Supplies the phenomenological and temporal layer: felt reorganization can precede naming, and persistence must be separated from momentary intensity.
Readout Genesis / MEMK	Retained difference precedes semantic naming; meaning follows readout; data, meaning, truth, and epistemic status remain non-identical; approximate translation must expose loss.	Prevents signal or sound from being treated as a container of finished meaning and prevents resonance from becoming a root ontology.
Epistemic Fusion or Tunnel	Articulation is a bounded readout of a larger human state; trajectories are reader-, context-, and history-sensitive; accessibility, retention, warrant, and direction do not collapse.	Supplies transport, reader-dependence, history-shaped accessibility, and non-collapse discipline.
CTSA Human-Return Readout	Explicit Human Return can be typed by what remains demonstrably usable after AI removal, while gain, loss, warrant, regulation, and direction remain separate.	Supplies a useful upper boundary: the present paper concerns a lower, not necessarily explicit, affective-semantic layer that may later become conceptual or skillful Human Return.

treat x_n as meaning itself.

Let $\mathcal{R}_H$ denote the human reader under current organization H_n and context c_n. A bounded readout is

$$r_n = \mathcal{R}_H(x_n \mid H_n, c_n).$$

The readout is not a copy of the source event. It is what becomes accessible to this reader under these conditions.

A context-sensitive affective-semantic configuration is then written schematically as

$$\mu_n = G(r_n, M_{n-1}, A_{n-1}, L_{n-1}, V_{n-1}, c_n),$$

where M denotes retained memory organization, A affective weighting, L currently available linguistic/conceptual organization, and V value/action relevance. The notation is architectural, not a claim that these objects are literal neural variables.

Lived experience is represented at a higher descriptive level as

$$E_n = \Phi(r_n, \mu_n, B_n, c_n),$$

where B_n marks embodiment. This inherits the level separation in Human LoRA: the lived event cannot be identified with a parameter update or a bare signal [15].

Only some residue is retained:

$$H_{n+1} = U(H_n, \text{Retain}(E_n, r_n, \mu_n), \delta^{world}_{n:n+1}).$$

Here δ^{world} represents later correction or resistance. An intense moment may therefore produce little durable change, while a quieter event may reorganize later interpretation.

Finally, explicit naming is optional and temporally placed after the readout event:

$$\ell_n = \mathcal{L}_H(E_n, \mu_n \mid H_n, c_n),$$

where ℓ_n is a linguistic articulation, label, metaphor, description, or question. The architecture therefore permits

$$E_n, \mu_n \quad \text{before} \quad \ell_n.$$

This is the formal sense in which meaning may precede naming.

6. Resonance Is Reorganization, Not Transmission

The everyday word *resonance* is useful only if stripped of literal physical import. The present proposal does not claim that human meaning behaves like coupled oscillators. Resonance is a reader-relative structural diagnostic for a case in which an incoming pattern makes some relations more accessible, more strongly weighted, or newly connected in the current human organization.

Let $\mathcal{A}_H^{\text{pre}}$ be a task- or experience-relative accessibility profile before an encounter and $\mathcal{A}_H^{\text{post}}$ the profile after it. A finite diagnostic may be written as

$$\Delta\mathcal{A}_H(x, c) = \mathcal{A}_H^{\text{post}} - \mathcal{A}_H^{\text{pre}}.$$

A positive change in one region does not imply global improvement. It means only that some relation, memory, possibility, or action tendency became more accessible under the declared reader.

Resonance without identity. Two people can be affected by the same external pattern without receiving the same meaning. The shared source constrains the encounter, but the readout is conditioned by different histories, contexts, bodies, memories, and conceptual repertoires.

Resonance without truth. A pattern can produce a powerful, coherent, memorable reorganization while the interpretation attached to it is false, ethically harmful, or epistemically unsupported. Affective force cannot substitute for warrant.

Resonance without retention. A strong momentary response may leave no durable reorganization. Conversely, a subtle encounter may later prove consequential if its residue is retained and integrated.

These propositions extend the programme's non-collapse discipline rather than adding a new ontology.

7. Language Is a Special Transport Layer, Not the Container of Meaning

The programme's earlier language model remains intact. A human state $Z_{H,t}$ is articulated through language as a bounded question or expression $Q_{H,t}$:

$$Z_{H,t} \xrightarrow{\mathcal{L}_H} Q_{H,t}.$$

Fusion/Tunnel emphasizes that $Q_{H,t}$ is not identical to the whole state; articulation exports only part of what is retained and available [17]. Once externalized, the expression can be compared, challenged, transported, and revised.

The present paper adds a prior possibility:

$$\begin{aligned}\text{pattern encounter} &\rightarrow \text{affective-semantic reorganization}\\ &\rightarrow \text{optional articulation}.\end{aligned}$$

Language therefore performs at least three distinct functions.

First, it can **name** a distinction that was already experientially active but tacit or only partially discriminated. Polanyi's account of tacit knowing provides a classical ally for the idea that explicit statement need not exhaust what is operative in human understanding [12].

Second, language can **restructure** experience. A later word, metaphor, diagnosis, story, or theoretical concept can alter which relations become salient and how an earlier event is remembered or interpreted. Human Learning's meaning maps and concept graphs operate precisely at this level [14].

Third, language can **transport** a bounded readout to another person or system. But transport necessarily introduces selection and loss. Bender and Koller distinguish linguistic form from meaning, while Bisk et al. emphasize the grounding problem for language detached from situated experience [2, 3]. These external literatures support the programme's refusal to identify text with the world or with the total lived state.

Thus the revised ordering is not "meaning without language" versus "meaning through language." It is layered:

$$\begin{aligned}&\text{readout} \rightarrow \text{lived significance}\\ &\leftrightarrow \text{language-mediated reorganization and transport}.\end{aligned}$$

8. From Felt Change to Retained Human Change

Human LoRA gives the temporal rule required here: experience is not automatically learning. An encounter may be vivid and vanish; another may be selectively retained, consolidated, and integrated into future dispositions [15, 6, 7, 10].

This matters because the intuitive language of "a song stayed with me," "a phrase changed me," or "a moment still echoes" can hide several different regimes:

- **Transient activation:** a relation becomes accessible during the encounter but returns near baseline afterward.
- **Recallable residue:** the event can later be remembered without materially changing interpretation or action.
- **Retained sensitivity:** later events are noticed, weighted, or discriminated differently.
- **Conceptual crystallization:** an initially difficult-to-name experience later becomes articulable as a concept or distinction.
- **Action reorganization:** the retained change affects later choice, avoidance, approach, practice, or inquiry.

The architecture therefore separates

$$\begin{aligned}\text{Exposure} &\neq \text{felt intensity}\\ &\neq \text{retention} \neq \text{improvement}.\end{aligned}$$

Karpicke and Roediger's work on retrieval and long-term retention and the broader consolidation literature provide external constraints on the distinction between momentary availability and durable learning [9, 6].

9. Relation to Human Return and CTSA

CTSA Human-Return Readout asks what capability remains demonstrably usable after AI support is removed. Its four readout classes—Conceptual, Tool-Selection, Skill-Execution, and Alternative-Generation Return—are deliberately explicit and behaviorally demonstrable [19].

The present architecture sits lower in the stack. An affective-semantic reorganization may be real and retained before it becomes an explicit concept, tool-selection rule, executable skill, or articulable alternative. This does not refute CTSA; CTSA already rejects the claim that it exhausts human cognition or all human learning. Instead, the present note clarifies a boundary condition:

> *Not every retained human change is immediately an explicit epistemic capability, but explicit Human Return may emerge from a prior affective-semantic reorganization that later becomes nameable, discriminable, and usable.*

A possible trajectory is therefore

$$\text{felt reorganization} \rightarrow \text{retained sensitivity} \rightarrow \text{naming} \rightarrow \text{Concept Return} \rightarrow \text{later Skill/Alternative Return}.$$

This is an open developmental pathway, not a universal causal sequence.

The reverse is also possible. A new concept supplied by language can reorganize later affective experience. A person who acquires a distinction may subsequently hear, see, or remember the same event differently. Human learning is therefore recursive rather than strictly bottom-up.

10. Context Wisdom as Governance of Resonant Meaning

The programme's Context Wisdom line treats wisdom not as possession of many meanings but as governance of meaning under context, provenance, resistance, and time. The present extension sharpens this point.

Affectively powerful meaning can become load-bearing very quickly. Repetition, prestige, ritual, group identity, aesthetic intensity, personal memory, or emotional relief may make a relation easier to re-enter. None of these facts alone establish its truth or ethical legitimacy.

Context Wisdom therefore asks at least four questions:

1. What did this encounter make newly accessible?
2. Which prior memories, values, identities, or expectations gave the pattern its current force?
3. What alternative readouts remain possible for other readers or in other contexts?
4. Which parts of the retained interpretation can still face independent resistance and revision?

The goal is not to suppress affect. It is to keep affective-semantic reorganization revisable rather than allowing intensity, familiarity, or repetition to become self-certifying.

11. Non-Collapse Laws

The extension is primarily a non-collapse architecture. The following distinctions are load-bearing:

$$\begin{aligned}
\text{external pattern} &\neq \text{meaning},\\
\text{meaning} &\neq \text{explicit naming},\\
\text{shared stimulus} &\neq \text{shared inner state},\\
\text{affective intensity} &\neq \text{epistemic warrant},\\
\text{accessibility} &\neq \text{truth},\\
\text{resonance} &\neq \text{understanding},\\
\text{retention} &\neq \text{benefit},\\
\text{verbalization} &\neq \text{lossless translation},\\
\text{same surface form} &\neq \text{same reader-relative meaning}.
\end{aligned}$$

Readout Genesis supplies the deepest guardrail: data are not meaning, meaning is not truth, and truth is not epistemic status [16]. The present note simply extends those separations into a domain in which felt significance can precede explicit naming.

12. Failure Modes and Adversarial Cases

12.1 The encoded-emotion fallacy

A listener reports sadness after an acoustic pattern. It does not follow that "sadness" existed as a semantic object inside the sound and was transferred intact. The sound constrained the readout; the reported sadness is a reader-relative interpretation of the encounter.

12.2 The universality shortcut

Several listeners respond similarly. Shared response does not by itself establish a context-free universal code. Common embodiment, enculturation, exposure history, task framing, or social expectation may contribute. The present paper makes no universality claim.

12.3 The intensity-as-truth error

An experience feels profound and memorable. That gives evidence about its force for the reader, not about the truth of every proposition later associated with it.

12.4 The naming-creates-meaning error

A person cannot initially describe an experience. It does not follow that there was no meaningful organization before the label appeared. Conversely, a later label can reshape the earlier experience; naming is neither mere reporting nor absolute creation.

12.5 The retention-is-good error

A pattern becomes deeply retained. Persistence can support learning, identity, care, or skill, but it can also stabilize fear, prejudice, fixation, or maladaptive expectation. Human LoRA and Fusion/Tunnel both require direction to remain separate from retention magnitude [15, 18].

12.6 The literal-resonance error

The term resonance is interpreted as a claim of shared oscillatory physics between minds. This is outside the present architecture. Resonance is only a structural shorthand for reader-relative reweighting, accessibility, and retained reorganization.

13. Empirical Programme

The first test should not attempt to validate a grand theory of meaning. It should test whether a prenominative readout layer adds explanatory value beyond simpler alternatives.

A minimal study could compare responses to matched nonverbal or weakly verbal patterns under different contexts. The design should freeze a pre-exposure baseline, record immediate experience without forcing premature verbal explanation, then obtain later articulation and delayed measures of retention or transfer. The key question is whether measurable reorganization appears before explicit naming and whether later naming predicts, merely reports, or further restructures the earlier change.

Candidate observables include:

- changes in association structure or accessibility before and after exposure;
- affective discrimination that precedes stable verbal labeling;
- delayed reconstruction of the experience and its boundaries;
- changes in later interpretation of related but novel patterns;
- divergence across readers exposed to the same stimulus under different histories or contexts;
- loss introduced by verbal translation: distinctions present in immediate experience but absent from later description;
- persistence versus decay across delayed measurement.

Several competing explanations must be preserved. Apparent "meaning before naming" could reduce to undifferentiated arousal, demand characteristics, post-hoc verbal reconstruction, memory bias, or familiarity. A strong design must include conditions capable of separating these alternatives.

H1 - Pre-nominative reorganization [Open]. For at least some classes of experience, task-relevant changes in affective or relational accessibility will be detectable before stable explicit naming.

H2 - Reader dependence [Open]. The same external pattern will produce systematically different readout profiles as reader history, context, or prior organization changes, even when the source pattern is held fixed.

H3 - Translation loss [Open]. Later verbal description will fail to preserve some distinctions present in immediate experience, while also adding distinctions that were not previously explicit.

H4 - Retention selectivity [Open]. Immediate intensity will not perfectly predict delayed retained reorganization.

H5 - Recursive naming [Open]. Naming after an initially difficult-to-articulate experience will sometimes reorganize subsequent readout of the same or related patterns rather than merely describe a fixed prior state.

H6 - No automatic improvement [Derived guardrail]. Even when H1–H5 are supported, no inference to truth, wisdom, well-being, or ethical value follows without independent criteria.

14. Legitimate Defeaters

The architecture should be narrowed or abandoned if any of the following occurs under well-specified tests:

- pre-nominative measures add no explanatory or predictive information beyond arousal and explicit labeling;
- reader history and context do not materially alter the relevant readout once stimulus features are controlled;
- the proposed distinction between immediate experience and later articulation cannot be operationalized reliably;
- delayed retained change is fully predicted by simpler memory variables with no need for an affective-semantic layer;
- "resonance" proves to be dispensable vocabulary that adds no measurable structure beyond accessibility or association change;
- the architecture cannot distinguish genuine retained reorganization from post-hoc narrative reconstruction.

These are removal conditions, not weaknesses to be hidden. The programme's methodological rule remains that a variable which performs no distinct explanatory work should be deleted.

15. Claim Boundaries

This paper does **not** claim that:

- music has one universal meaning or emotional code;
- specific musical features universally cause specific emotions;
- the inner state of a performer is transferred intact into a listener;
- all meaningful experience is pre-linguistic or independent of language;
- affective-semantic reorganization is a literal neural module;
- resonance is literal physical resonance between minds;
- a strong emotional response establishes truth, authenticity, morality, or spiritual validity;
- every retained experience improves the person;
- the equations are validated psychological or neuroscientific laws;
- CTSA, Human Return, or the present layer exhaust all forms of human learning;
- self-citation from the programme constitutes independent empirical support.

A dedicated music-cognition paper would require a separate review of auditory perception, predictive timing, tonal expectation, entrainment, affect, memory, culture, and individual differences. Those claims are deliberately outside the present note.

16. Standalone Thesis

> *Meaning need not begin with a name. A finite human can undergo an affectively weighted, context- and history-dependent reorganization before that change is propositionally articulated. External patterns do not transport finished meanings from one interior to another; they constrain a reader-relative readout whose significance is co-constituted by embodiment, memory, expectation, language, value, and context. Some reorganizations vanish, some are later named, and some are selectively retained so that future situations are read differently. Music makes this architecture visible, but does not define it.*

The programme sequence can now be written as

$$\text{retained difference} \to \text{bounded readout} \\ \to \text{lived affective-semantic configuration} \\ \to \text{optional naming} \to \text{retention/reorganization} \\ \to \text{altered future readout}.$$

This extension preserves the earlier architecture while moving one level beneath language. Human Learning explains how named meaning becomes corrigible competence; Human LoRA explains how lived events can leave durable residues; Readout Genesis prevents semantic labels from becoming root primitives; Fusion/Tunnel explains how articulation becomes transport and how history shapes later accessibility; CTSA measures some explicit forms of what later returns as capability. The present note adds the missing phenomenological bridge: **what can become meaningful in the human before it can yet be fully said.**

17. References

[1] Anderson, J. R. (1982). Acquisition of cognitive skill. *Psychological Review, 89*(4), 369–406.

[2] Bender, E. M., & Koller, A. (2020). Climbing towards NLU: On meaning, form, and understanding in the age of data. *Proceedings of ACL 2020*, 5185–5198. https://doi.org/10.18653/v1/2020.acl-main.463

[3] Bisk, Y., Holtzman, A., Thomason, J., Andreas, J., Bengio, Y., Chai, J., et al. (2020). Experience grounds language. *Proceedings of EMNLP 2020*, 8718–8735. https://doi.org/10.18653/v1/2020.emnlp-main.703

[4] Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences, 36*(3), 181–204. https://doi.org/10.1017/S0140525X12000477

[5] Collins, A. M., & Loftus, E. F. (1975). A spreading-activation theory of semantic processing. *Psychological Review, 82*(6), 407–428.

[6] Dudai, Y. (2004). The neurobiology of consolidations, or, how stable is the engram? *Annual Review of Psychology, 55*, 51–86. https://doi.org/10.1146/annurev.psych.55.090902.142050

[7] Dudai, Y. (2012). The restless engram: Consolidations never end. *Annual Review of Neuroscience, 35*, 227–247. https://doi.org/10.1146/annurev-neuro-062111-150500

[8] Fillmore, C. J. (1982). Frame semantics. In Linguistic Society of Korea (Ed.), *Linguistics in the Morning Calm* (pp. 111–137). Hanshin.

[9] Karpicke, J. D., & Roediger, H. L. (2007). Repeated retrieval during learning is the key to long-term retention. *Journal of Memory and Language, 57*(2), 151–162. https://doi.org/10.1016/j.jml.2006.09.004

[10] McGaugh, J. L. (2015). Consolidating memories. *Annual Review of Psychology, 66*, 1–24. https://doi.org/10.1146/annurev-psych-010814-014954

[11] Noe, A. (2004). *Action in Perception*. MIT Press.

[12] Polanyi, M. (1966). *The Tacit Dimension*. University of Chicago Press.

[13] Varela, F. J., Thompson, E., & Rosch, E. (1991). *The Embodied Mind*. MIT Press.

[14] Lahtee, Y. (2026). *Human Learning as Epistemic Architecture: A Method for Word Mapping, Life-Concept Graphs, Constraint Testing, and Corrigible Agency in the AI Age*. Revised v4 preprint, 28 June 2026.

[15] Lahtee, Y. (2026). *Experience Is the Human LoRA: A Readout-Retention Theory of Selective Model Change*. Open Civil Science Initiative preprint, 18 July 2026.

[16] Readout Genesis Programme. (2026). *READOUT_GENESIS_MEMK_DOMAIN v0.3.0 ANCHORED*. Root-connected semantic-cognitive domain specification, 23 July 2026. Architecture-conditional / unresolved domain.

[17] Lahtee, Y. (2026). *Epistemic Fusion or Tunnel: Conversation-Centered Architecture*. Human-AI Readout Programme, Conversation-Centered v3 FINAL, 5 September 2026.

[18] Lahtee, Y. (2026). *Epistemic Fusion or Epistemic Tunnel: A Phenomenology-Anchored, Global-Literature-Constrained, Problem-First Architecture of Human-AI Collaboration*. Human-AI Readout Programme, v8.1 strong-claim / legitimate-defeater edition, 5 September 2026.

[19] Lahtee, Y. (2026). *CTSA Human-Return Readout: A Session-Boundary Measurement Architecture for Retained Human Capability After AI*. Human-AI Readout Programme, v9.1 FINAL / global-literature-constrained edition, 5 September 2026.